Natural Capital and the Measurement of Productivity*

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Abstract

Omitting inputs to production, such as natural capital, affects estimates of productivity growth. We show benchmark measurements of productivity growth are strongly correlated with environmental change at the national level, consistent with biased measurement due to the omission of natural capital. We derive theoretical conditions under which the measurement and inclusion of previously omitted environmental inputs reduce bias and error when estimating productivity growth. Calibrated simulations using data from the World Bank and Penn World Tables suggest including non-renewable natural capital in the measurement of TFP growth increases accuracy for 96 of the 100 countries in our sample.

Keywords: Total Factor Productivity; Natural Capital; Economic Measurement

JEL Codes: E01, O44, O47, Q56

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Innovation and technological change drove an unprecedented increase in economic activity over the past two centuries. At the same time, industrialization muddled societal awareness of fundamental connections between humans and nature. Society's interest in tabulating the stocks of nature and their associated flows waned, and attention shifted to tracking produced and human capital. Then-President Theodore Roosevelt noted this phenomenon at the outset of the 20$^{\text{th}}$ century:

> "With the rise of peoples from savagery to civilization, and with the consequent growth in the extent and variety of the needs of the average man, there comes a steadily increasing growth of the amount demanded by this average man from the actual resources of the country. And yet, rather curiously, at the same time that there comes that increase in what the average man demands from the resources, he is apt to grow to lose the sense of his dependence upon nature." – Roosevelt (1908)

Formal systems of economic measurement followed this trend; extant structures provide little guidance for separating nature's contribution to production from the contributions of other factors. Starting with Simon Kuznets' report to Congress in 1934, the broader development of statistical, economic, and environmental policy has focused on measuring output and productivity growth by tracking capital and labor over time (Kuznets 1934). While governments historically emphasized tracking natural resources for posterity and avoiding overuse, features of the environment beyond land were omitted from the transition to the double entry-based national accounting systems used today. For example, sailors, vessels, and sonar are essential inputs to industrial fishing and are measured by national accountants on a regular basis; equally important is the fish stock and its ability to replenish itself over time (Hannesson 2007). However, the fish stock as an input to production is often excluded from economic measurement. This continues despite managed fish stocks (within exclusive economic zones) being a textbook example of a non-produced natural asset scheduled for measurement under modern standards like the System of National Accounts (United Nations 2009, 2025).

Here, we examine how the omission of the environment as a factor in production affects a leading measure of economic progress produced under current national accounting standards, total factor productivity (TFP). TFP growth is defined as the growth in output unexplained by growth in inputs, and it has been a benchmark metric for economy-wide technological progress and innovation for nearly 70 years (Solow 1957). Changes in TFP over time play a crucial role in shaping governments' budget projections and policy agendas over long time horizons (CBO 2025). If natural capital affects production, then the omission of environmental assets from national accounting frameworks induces error when estimating aggregate productivity growth. Failing to account for how changes in the environment affect output may lead governments to overestimate the ability of society to grow without the use of nature or to misattribute the effects to economic policy (Nordhaus and Tobin 1973; Brandt, Schreyer, and Zipperer 2017).

In this paper, we first demonstrate how omitting inputs from the production function affects residual-based estimates of TFP and discuss why natural capital is often omitted in practice. We provide predictions for the correlation of environmental change with TFP growth when natural capital is incorrectly omitted from the set of measured inputs used to construct TFP. Then, we test whether a benchmark series for TFP growth is orthogonal to proxies for natural services in a balanced panel of data for 117 countries between 1996 and 2019. We find measured growth is strongly correlated with changes in the modest set of natural capital stocks we consider. The correlations between productivity growth and natural capital persist when using alternative productivity measurements from the economics literature or those provided by national statistical agencies.

Next, we characterize when inclusion of natural capital aids measurement of TFP in theory and consider whether it does so in practice. We derive conditions under which additional measurement unambiguously reduces bias in residual-based estimates of TFP growth. These conditions prove to be restrictive and unlikely to hold in most settings. As the theoretical conditions are not formally testable, we use simulation-based evidence to gauge whether including natural capital is likely to improve measurement empirically. The results suggest that including natural capital produces gains in measurement accuracy. Across simulations calibrated to data from 100 countries, the median level of increase in accuracy, measured as the reduction in root mean squared error for estimated average TFP growth, is 6.6 basis points, or a little over 14 percent of the average TFP growth across all countries during the sample period. For the case of the United States, compounding an error reduction of this magnitude over a standard 10-year budget horizon implies a forecast error for cumulative revenues of a little over $300 billion in 2025. This figure is two to three times larger than the annual year-on-year forecast error in deficit projections by the U.S. Congressional Budget Office (CBO), or roughly a quarter of the aggregate cost of compliance with the Clean Air Act between 1970 and 1990 (CBO 2026; U.S. Environmental Protection Agency 1997).

Literature: Tobin and Nordhaus motivated the inclusion of natural capital when accounting for growth by appealing to its amenity value (toward their broader measure of economic welfare) as well as the productive value of land (Nordhaus and Tobin 1973). Subsequent research incorporating the environment in the development accounting literature has primarily focused on improving measures of the share of national income attributable to produced capital (e.g., Mankiw, Romer, and Weil 1992; Caselli and Feyrer 2007; Monge-Naranjo, Sánchez, and Santaeulalia-Llopis 2019). These studies apportion capital income between returns on investment in produced capital and resource rents. Broadly, the literature has highlighted the importance of correcting for resource rents when comparing the returns to capital across countries while noting that the amount of residual output unexplained by factor inputs remains substantial (Hsieh and Klenow 2010). We add to this literature by characterizing how introducing previously omitted factors of production has ambiguous effects on the bias of measured output elasticities and factor shares.

A more nascent strain of literature focuses on how including natural capital affects growth accounting-based estimates of productivity. Our contribution is closest to those by Brandt, Schreyer, and Zipperer (2017) and Manning and Barbier (2025), who study the effects of adjusting existing TFP estimates at the national level to account for the extraction of subsurface minerals and fossil fuels. Related work by Freeman, Inklaar, and Diewert (2021) highlights the importance of accounting for the fact that not all countries extract the same set of non-renewable resources domestically when comparing productivity levels across countries. This is distinct from our exercise comparing productivity growth within countries across time, but presents a methodology that would allow our calibration exercise to be applied to more granular measures of TFP. We add to this literature by explicitly characterizing the conditions under which measurement of non-renewable resources reduces errors in estimated TFP, as opposed to calculating the magnitude of the adjustments induced by inclusion. Our approach can also be generalized to consider the effects of including other potentially omitted capital stocks, such as renewable natural capital.

1 Residual-Based Productivity Measurement

Neoclassical economics attributes output to two sources: factors of production (factor inputs) and technology. Final output $Y(t)$ is determined by the level of factor inputs, the vector $\boldsymbol{X}(t)$, and a summary statistic for the level of technology available to society, total factor productivity, $TFP(t)$, each instant:

$$Y(t) = TFP(t) \times F(\boldsymbol{X}(t)). \tag{1}$$

TFP is a theoretical modeling device. It is an index number capturing how technology determines the production possibility frontier at a given point in time.[1] It is a parsimonious treatment for how forces such as innovation and creative destruction shift what society can produce given available resources (Romer 1990; Aghion and Howitt 1992). Under standard assumptions for the production function (e.g., the Cobb-Douglas or transcendental logarithmic forms), growth in output is given by

$$g_Y(t) = g_{TFP}(t) + \sum_{x \in \boldsymbol{X}(t)} \gamma_x(t) g_x(t), \tag{2}$$

where $g_x(t)$ terms denote growth rates and $\gamma_x(t)$ are the output elasticities for each input x at time t.[2] The growth accounting equation in (2) states that all changes in output are attributable to factors of production or growth in TFP (Barro 1999). Rearranging the growth accounting equation gives

$$g_{TFP}(t) \triangleq g_Y(t) - \sum_{x \in \boldsymbol{X}(t)} \gamma_x g_x(t); \tag{3}$$

TFP growth is the output growth unexplained by growth in inputs (Comin 2010). The accounting identity in (2) requires that these two concepts of productivity growth—the growth rate of an index number for technology and the residual portion of growth unexplained by factor inputs—coincide.

1. We assume throughout this paper that technical change is Hicks neutral. A more general formulation of technical change is the partial derivative of the function determining aggregate output with respect to time.

2. Here we refer to the $\gamma_x(t)$ terms as output elasticities. Section 1 of the online appendix shows that all analysis holds when the $\gamma_x(t)$ terms are reinterpreted as the income shares for factors of production.

In practice, TFP growth is measured using the portion of output growth not explained by growth in a set of *observed* factor inputs, which we group together as the set $\boldsymbol{I}$. Factor inputs (e.g., produced capital and labor) and final output are measured by national statistical agencies at regular frequencies, allowing for statisticians to take versions of (3) to the data. Residual changes in observed output—those unexplained by growth in observed inputs—are called the Solow residuals (Solow 1957) and are imputed to productivity growth by solving

$$g_A(t) \triangleq \hat{g}_Y(t) - \sum_{i \in \boldsymbol{I}} \hat{\gamma}_i(t)\hat{g}_i(t), \tag{4}$$

where $\hat{g}_i$ and $\hat{\gamma}_i$ are estimates of growth and output elasticities, respectively. A Solow residual is an estimate of TFP growth. Versions of equation (4) are used by national statistical agencies to construct productivity growth in national accounts (Meyer and Harper 2005; European Commission and Eurostat 2025), and in academic applications comparing productivity across countries and over time (Feenstra, Inklaar, and Timmer 2015). Statistical agencies often label their versions of the estimand in (4) *multifactor productivity* (as opposed to TFP) to acknowledge that they measure productivity growth beyond the factors they include (Organisation for Economic Co-operation and Development 2001).

If national accountants were able to calculate the dynamics of all potential factor inputs, a natural candidate for $\boldsymbol{I}$ would be to include all observed stocks to ensure $\boldsymbol{X} \subseteq \boldsymbol{I}$ (Nordhaus and Kokkelenberg 1999). In the interim, the set of measured inputs $\boldsymbol{I}$ available to practitioners remains limited. Current residual-based approaches to productivity measurement deployed by national statistical agencies and researchers do not include environmental inputs (Meyer and Harper 2005; Feenstra, Inklaar, and Timmer 2015; European Commission and Eurostat 2025; United Nations 2025). These omissions are not active decisions to ignore nature; instead, they reflect limited state capacity and the difficulty of measuring how natural capital is used in production. The service flows from natural capital that enter the production function often go well beyond the simple accounting of resources that are extracted (e.g., fish stocks and subsurface oil pressure). Measurement is further complicated by the difficulty of attributing factor income to non-market and public environmental services (e.g., precipitation and climate).

To illustrate the effects of omitting natural capital inputs when estimating TFP, consider an economy with four factor inputs: produced capital, labor, non-renewable natural capital, and renewable natural capital, such that $\boldsymbol{X}(t) = \{K(t), L(t), N_1(t), N_2(t)\}$. For this economy, if a national accountant includes only produced capital and labor as inputs $\boldsymbol{I}$ when estimating TFP, conditional on her estimates of factor input growth rates and output elasticities $(\hat{\boldsymbol{g}}, \hat{\boldsymbol{\gamma}})$ the Solow residual can be decomposed as

$$g_A(t)|\boldsymbol{I}, \hat{\boldsymbol{\gamma}}, \hat{\boldsymbol{g}} = \underbrace{g_{TFP}(t)}_{\text{True innovation}} + \underbrace{\sum_{x \in \{N_1, N_2\}} \gamma_x(t) g_x(t)}_{\text{Omitted inputs}} + \underbrace{\sum_{i \in K,L} \left[\gamma_i(t)g_i(t) - \hat{\gamma}_i(t)\hat{g}_i(t)\right]}_{\text{Measurement error}}. \tag{5}$$

When natural capital inputs are omitted and $\boldsymbol{X} \setminus \boldsymbol{I} \neq \emptyset$, (3) is a biased estimator for TFP growth; the contributions of the omitted natural capital inputs are misattributed to TFP because they are spuriously absorbed by the Solow residual.[3] Correlation of the Solow residual in (5) with changes in a given natural capital stock (or other potentially-omitted input) over time is evidence that the stock in question may be an omitted input in production. Conversely, a precise null of zero correlation between a natural capital stock and estimated TFP growth suggests that the empirical growth accounting equation contains the true set of factor inputs (i.e., $\boldsymbol{X} \subseteq \boldsymbol{I}$), and omitting the candidate stock for inclusion is unlikely to bias measures of TFP growth.

1.1 Testing for Omitted Natural Capital Inputs

The decomposition in (5) shows three channels by which residual-based estimates of TFP growth can be correlated with natural capital: (1) correlation between natural capital use and unobservable growth in TFP, (2) correlation with omitted inputs that affect output (e.g., natural capital stocks that are elements of $\boldsymbol{X}$ and omitted when estimating TFP growth), and (3) correlation with measurement error when calculating growth rates or output elasticities of included factors. While an econometrician cannot distinguish among these three potential sources of correlation, she can test whether the Solow residuals are uncorrelated with changes in natural capital stocks. If growth in natural capital is correlated with measured TFP growth, then we cannot rule out channel (2). Finding this form of empirical correlation in the data suggests that natural capital is more likely to play a role in output.

In a world with perfect information over the true growth rates of factor inputs and TFP, we could falsify that a given natural capital stock is an omitted variable (channel 2). However, as we only observe g_A jointly as the sum of the three terms in (5), we are restricted to testing whether the correlations between growth in natural capital stocks and *measured* TFP are nonzero. This leads to the potential for a form of type one error where we correctly reject the null hypothesis of no correlation between g_A and g_{N_i}, but spuriously attribute it to N_i being an omitted input in production instead of some combination of channels 1 (correlation with unobservable TFP growth), 2 (correlation with other omitted inputs), or 3 (correlation with measurement errors). This also brings about the potential for a form of type two error: when g_{N_i} is an omitted variable in the growth accounting equation and correlated with measurement error and/or unobservable TFP growth, these correlations may offset and lead the econometrician to (correctly) fail to reject the null hypothesis of no correlation between g_A and g_{N_i}, but spuriously attribute this to natural capital playing no role in production.[4]

3. It is possible measurement error exactly cancels out the omitted terms and the estimator is unbiased even when inputs are omitted. We view this case as unlikely.

4. Several features of measurement may bring about these spurious results in practice. First, estimating output elasticities is nontrivial even for cases where factor income can be reasonably attributed (Caselli and Feyrer 2007). Second, the lack of attributed factor income makes it difficult to delineate which changes in measured natural capital stocks constitute proxies for changes in flows entering production. Finally, while measures of some forms of innovation exist (e.g., patent filings and seed varietals, used as proxies for innovation in Acemoglu et al. 2023 and Moscona and Sastry 2023 respectively), underlying correlations between environmental change and TFP growth are not falsifiable.

1.2 Estimating Equation

To test for correlations between natural capital and measured productivity, we regress measured national TFP growth ($g_{A_{j,t}}$) from the Penn World Tables (PWT) version 10.01 (Feenstra, Inklaar, and Timmer 2015) on growth in the seven country-level natural capital stocks ($g_{N_{n,j,t}}$) measured by the 2024 version of the World Bank's The Changing Wealth of Nations (CWON) dataset. Equation (6) shows the empirical analog of (5) that we take to the available data. We estimate (6) on a balanced panel of 117 countries over the 1996-2019 period:[5]

$$g_{A_{j,t}} = \sum_{i \in \boldsymbol{I}} \beta_i g_{N_{i,j,t}} + \sum_{z \in \boldsymbol{Z}} \theta_z g_{z,j,t} + \delta_j + \zeta_t + \varepsilon_{j,t} \tag{6}$$

where j and t index countries and years respectively. The set $\boldsymbol{I}$ covers the seven natural capital stocks measured at the country level present in the CWON, all measured in physical quantity units: urban land, agricultural land, timber land, non-timber forests, mangroves, fishery biomass, and hydroelectric power generation.[6] Coefficients β_i govern their relationship with TFP growth. The summation over terms $g_{z,j,t}$ allows for specifications that control for growth in other factor inputs such as produced capital, employment, labor's share of income, and human capital taken directly from the PWT. Terms (δ_j, ζ_t) allow for country and year fixed effects (labeled TWFE in Table 1), and $\varepsilon_{j,t}$ are error terms capturing residual idiosyncratic variation in TFP growth.

The β_i terms are the estimands of interest. While (6) resembles (5), the β_i coefficients on natural capital growth rates are not the associated output elasticities. Inference on these coefficients does not test for causal effects from changes in natural capital stocks on productivity growth, nor do these coefficients separately identify the omitted variable bias we seek to test from other potential channels for correlation. Rather, these coefficients identify the (conditional) covariance between growth in natural capital stocks and growth in measured TFP. Failing to reject the joint null hypothesis of interest:

$$H_0 : \boldsymbol{\beta} = \boldsymbol{0} \tag{7}$$

only indicates that we cannot rule out that the set of natural capital stocks we consider is uncorrelated with the combination of components in the decomposition of TFP growth in (5).

Three assumptions must hold in order to interpret the rejection of the null hypothesis in (7) as evidence that natural capital inputs are omitted when calculating TFP. First, measurement error in the (weighted) factor input growth terms used to construct estimates of TFP is uncorrelated with growth in natural capital stocks. Second, changes in natural capital stocks are a good proxy

5. The CWON dataset in its raw form contains an unbalanced panel of natural capital stocks in each country covering 1995-2019. We impute missing values as zero changes from the last period where measurements were available, leaving us with a balanced panel of growth in natural capital stocks between 1996 and 2020. See Sections 2.2 and 2.3 of the online appendix for a discussion of the imputation process for missing data along with robustness checks to ensure the results are not an artifact of this process.

6. A full description of how the data on individual natural capital stocks are constructed in the CWON, how we clean these data prior to estimation, along with the list of countries covered in our analysis can be found in online appendix 2.2.

for changes in the flow of services they produce, and they are uncorrelated with any other omitted factor inputs beyond the natural capital stocks we consider. Third, the rate of true technical change g_{TFP} is conditionally uncorrelated with changes in environmental inputs. These assumptions are strong. Nevertheless, natural capital measurement remains important for national accountants even when they do not hold. Failures of the first or second assumptions imply that better environmental measurement improves either the accounting of existing factor inputs or the tracking of other economically relevant inputs, while failure of the third implies that natural capital measurement is essential for understanding how environmental change affects true productivity growth.

1.3 Reduced-Form Results

Table 1 displays estimates of $\hat{\beta}$ from various specifications of (6), how expected growth in TFP varies with changes in the growth of a given natural capital stock. Each column gives two tests of the null hypothesis in (7) that the vector of coefficients on growth in natural capital $\boldsymbol{\beta} = [\beta_1, \ldots, \beta_7]'$ is the zero vector. The bottom rows of table 1 show p-values from two Wald tests of this joint null hypothesis, one using asymptotic standard errors and a second using a semi-parametric wild bootstrap approach (Wu 1986). We can reject that the natural capital stocks we consider are uncorrelated with productivity growth at traditional significance levels under the tests using asymptotic standard errors across all specifications, and the tests using the wild bootstrap approach are always statistically significant at the 10 percent level. Almost all point estimates are positive, the expected sign if the natural capital stocks we consider are beneficial inputs for production.[7]

Robustness: Individual tests of coefficient precision suggest the services from land are the inputs most likely to affect measured TFP. To ensure missing land rents are not the sole drivers of the results here, we re-estimate equation (6) after excluding urban, agricultural, and timber land from the set of natural capital stocks we include as covariates. Under the nonland specification, our estimates lose precision and we are only able to reject the null hypothesis at the 10 percent level (see online appendix 2.3.2). This confirms the suggestion in Feenstra, Inklaar, and Timmer (2015) that better accounting for land rents in national accounts is a fruitful direction for future work. Online appendices 2.3.1 and 2.3.2 provide several additional checks of the robustness of our results to alternative choices for the set of natural capital stocks considered and confirm the results here. Online appendix 2.3.3 shows our results also hold when using alternative measures of productivity growth produced by government statistical agencies as outcome variables.

2 The Marginal Inclusion of Potentially Omitted Inputs

Section 1 provides suggestive evidence that natural capital is an omitted input in existing measures of TFP. We now turn to examining whether including previously omitted inputs is helpful for

7. The negative coefficient on forest cover may be due to increases in non-timber forest cover coinciding with declines in existing market activities making use of this land. This does not imply that forests are a drag on the economy, but instead that the services they produce are not captured by measured output.

measurement. Intuition suggests that including an omitted determinant of output in the growth accounting equation will improve estimates of TFP. However, this need not hold true. To build intuition for why this is the case, consider when the residual-based estimator is applied to our four-input example economy by a national accountant who measures growth in output, labor, produced capital, and non-renewables, but cannot measure growth in renewable resources over time (g_{N_2}). With full knowledge of the output elasticities for the first three factors in hand, if the accountant elects to omit non-renewables (g_{N_1}) when estimating TFP growth, substituting this into (4) yields the estimand

$$\hat{g}_A(t) \triangleq g_{TFP}(t) + \gamma_1 g_{N_1}(t) + \gamma_2 g_{N_2}(t) \tag{8}$$

If she instead includes non-renewables, the estimand for TFP growth is

$$\tilde{g}_A(t) \triangleq g_{TFP}(t) + \gamma_2 g_{N_2}(t) \tag{9}$$

It is ambiguous whether (9) is a less biased estimate of g_{TFP} than (8) without making assumptions on the sign of growth in renewable inputs over time. Section 1.4 of the online appendix shows the intuition here holds quite generally; including an omitted input on the margin reduces bias only when all omitted terms share a sign in expectation (Lemma 1). Necessary conditions for improving in-sample accuracy based on the root mean squared error (RMSE) criterion are even more stringent. Including natural capital on the margin, even with perfect measurement, is only guaranteed to reduce the RMSE of estimates of TFP when it leads the estimator to be correctly specified (i.e., $\boldsymbol{X} \subseteq \boldsymbol{I}$) or the covariance structures across all omitted inputs are very similar (Proposition 2).

3 Simulating the Effects of Incorporating Natural Capital when Estimating TFP

Section 2 examined the theoretical effects of reducing the number of omitted factor inputs on the performance of the Solow residual as an estimator of TFP. This section evaluates this question using the available data on growth in factor inputs, natural capital stocks, and output by taking our running four-input economy to the data.

3.1 Setup

We consider a case where a national accountant seeks to measure TFP growth for this economy, but measures only growth in the use of produced capital, labor, and non-renewable capital, N_1. We assume she suspects that services from renewable capital (N_2) are important inputs in production, but that she is unable to measure how their use has changed over time. Knowing this, she must choose whether to include growth in the use of non-renewable resources, g_{N_1}, when taking Equation (4) to the data to estimate TFP growth. Her criterion for including non-renewable resources in this thought experiment is whether it lowers expected in-sample root mean squared error when estimating average TFP growth in her data, which we assume covers 25 years of growth in output and factor inputs.

Data Generating Process for Output: Output growth in each country comprises growth in TFP, labor, produced capital, and two forms of natural capital inputs (non-renewables N_1 and renewables N_2). Joint observations of growth rates for all determinants of output growth are assumed to be independent and identically distributed draws of vectors $\boldsymbol{g}_{j,t}$ from a fixed distribution for each country with mean $\boldsymbol{\mu}_j$ and variance Σ_j. Output growth in each simulated country j is generated by

$$g_{Y,j,t} = \boldsymbol{g}'_{j,t}\boldsymbol{\gamma}_j \tag{10}$$

where $\boldsymbol{\gamma}_j \triangleq [1, \gamma_K, \gamma_L, \gamma_1, \gamma_2]'$ is the vector of output elasticities for country j.

Calibrating Country-Level Growth: We assume at the country level that growth in TFP (g_{TFP}) is uncorrelated with growth in other factors. We calibrate the mean ($\boldsymbol{\mu}_j$) and variance (Σ_j) governing the joint distribution of growth terms $\boldsymbol{g}_{j,t}$ in each country based on their analogs in the CWON and PWT. For natural capital, we follow the World Bank CWON methodology (World Bank 2025) and construct aggregate Törnqvist quantity indices for non-renewable and renewable resource use at the country level and set the parameters governing average growth in the use of natural capital equal to the sample means of growth in these indices between 1996 and 2019.[8] Expected growth in produced capital services, labor, and TFP are set equal to their sample analogs in the Penn World Tables over the same 1996-2019 period. Table 16 of the online appendix gives a full accounting of all data used in the simulation exercise.

Output Elasticities: We calibrate the output elasticity of services from renewable natural capital, γ_2, to equal 0.05 based on the analysis in Section 3.6 of the online appendix. In turn, to induce an overall output elasticity of 0.1 for natural capital (following Arrow et al. 2004), we set the output elasticity of services from non-renewable natural capital, γ_1, to 0.05. The sensitivity of our results to these choices of elasticities for natural capital is discussed below.

Procedure for Estimating TFP Growth: We assume the national accountant observes 25 years' worth of data on growth in factor inputs K, L, and N_1 and output Y in a given country. We assume she does not observe the distribution of national income earned across factors, and therefore cannot estimate output elasticities using factor shares. This ensures our thought experiment is consistent with estimating the Solow residual in empirical settings, where factor income generated

8. Section 3 of the online appendix outlines the full procedure for how we construct the Törnqvist indices for both forms of natural capital and calibrate $\boldsymbol{\mu}_j$ and Σ_j for each country.

by non-market natural capital services is unobservable to an econometrician (Manning, Taylor, and Wilen 2018).

To construct a measure of TFP growth, which does not rely on growth in non-renewable resources, the national accountant estimates:

$$g_Y = \hat{\gamma}_A + \hat{\gamma}_K g_K + \hat{\gamma}_L g_L + \epsilon \tag{11}$$

on time series of data for a given country using ordinary least squares (OLS). Her estimator for average TFP growth based only on growth in produced capital and labor is the intercept term $\hat{\gamma}_A$. We refer to this estimator as model 1. She then estimates the analog for the case including non-renewable resources:

$$g_Y = \tilde{\gamma}_A + \tilde{\gamma}_K g_K + \tilde{\gamma}_1 g_{N_1} + \tilde{\gamma}_L g_L + \nu \tag{12}$$

on time series for growth in produced capital, labor, and non-renewables. The OLS-based estimator for average TFP growth inclusive of non-renewable capital is $\tilde{\gamma}_A$. We refer to this estimator as model 2.

We then calculate the expected RMSE for $\hat{\gamma}_A$ and $\tilde{\gamma}_A$ in each country by evaluating models 1 and 2 at the sample analogs of $\boldsymbol{\mu}$ and Σ and comparing these estimates relative to the calibrated values, $\mathbb{E}[g_{TFP}]$:

$$\Delta\text{RMSE}|\boldsymbol{\mu}, \Sigma \triangleq \sqrt{\mathbb{E}[(\tilde{\gamma}_A - \mathbb{E}[g_{TFP}])^2|\boldsymbol{\mu}, \Sigma]} - \sqrt{\mathbb{E}[(\hat{\gamma}_A - \mathbb{E}[g_{TFP}])^2|\boldsymbol{\mu}, \Sigma]} \tag{13}$$

Applying this process to each country in the sample with sufficient data yields a cross section of 100 country-level observations of ΔRMSE.[9]

3.2 Results

For 96 out of the 100 countries for which we have sufficient data, including non-renewable resources (going from model 1 to model 2) lowers the RMSE of estimated TFP growth. Observations that fall in the green bins to the left of $x = 0$ in Figure 1 are countries where including non-renewable resources decreases the RMSE for a national accountant's residual-based estimate of TFP. The four observations in the blue bins to the right of $x = 0$ are countries for which including non-renewables increases the RMSE of estimated TFP growth (and decreases accuracy under this criterion). The median and mean declines in RMSE are sizable: reductions of 6.6 and 32 basis points (one-hundredth of a percentage point, abbreviated as bps) respectively. Reductions in error are dispersed (the standard deviation is 65 bps) and left-skewed. The countries with the largest reductions in RMSE are Kenya, Laos, Israel, Tanzania, and Niger. The magnitude of the median reduction is a little over 14 percent of average TFP growth across all countries over the 1996-2019 period. For the four countries where including non-renewables raises RMSE, the deterioration in fit

9. See Section 4.1 of the online appendix for the full procedure.

is small (less than 1 basis point both on average and at the median). The greatest RMSE increase occurs for Chile, where RMSE rises by 3.3 bps.

To decompose the contribution of individual components of output growth to changes in measurement accuracy, we regress ΔRMSE on the individual components of $\boldsymbol{\mu}_j$ and Σ_j. This gives a reduced-form sense of the elements of input growth most important in determining the effects of measuring non-renewables.[10] Online appendix Table 18 shows that ΔRMSE is decreasing in the country-level growth rates of renewable and non-renewable resource use, and increasing in the correlations between growth in non-renewable resource use and those for labor and produced capital.[11] Thus if the world looks increasingly like Spain or Kyrgyzstan—nations where growth in the use of non-renewables is negatively correlated with produced capital and labor, then including non-renewable resources in national accounts is more likely to reduce the error in estimates of TFP growth. In contrast, if the world becomes more like Chile or Indonesia, where growth in the use of non-renewables is positively correlated with produced capital and labor, then adding non-renewables to national accounts will have a more limited impact on the error in existing estimates of TFP growth.

Figure 2 shows scatter plots illustrating the cross-sectional variation in the relative performance of the estimators against individual country-level characteristics such as trends in resource use and baseline levels of calibrated productivity growth. Panels 2a and 2b plot changes in RMSE when estimating TFP growth against average growth in the use of non-renewable and renewable resources (g_{N_1} and g_{N_2}) at the country level. Of the 100 countries in our sample, 68 countries exhibit positive average growth in their use of non-renewables, and 47 exhibit positive average growth in the use of renewable inputs. Importantly, panels 2a and 2b show that average growth in non-renewables is about an order of magnitude greater than growth in renewables. This proportionally larger growth in non-renewable resources is responsible for substantially more output growth than that of renewables, even when the contribution of both resources to output in levels is similar. In turn, leaving out non-renewables on the margin leaves much more unexplained output in model 1 relative to model 2 where only renewables are omitted.[12]

Panels 2c and 2d show point estimates of TFP growth from models 1 and 2 respectively against the true data generating process (set as the observed value in the PWT). Panels 2e and 2f show the changes in RMSE from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this change. The median (mean) decrease in RMSE is 10 (22) percent of baseline error when both natural capital stocks are omitted, and rises to more than 40 percent for over a quarter of the countries in the cross section of simulated

10. The exact partial derivatives can also be calculated on a country-by-country basis using the closed-form solutions for ΔRMSE in Sections 4.2 and 4.3 of the online appendix.

11. Three of these four determinants of ΔRMSE are significant at traditional (5 percent) levels when ΔRMSE is regressed on the elements of $\boldsymbol{\mu}_j$ and Σ_j that enter the RMSE calculations. The joint test is significant at the 0.1 percent level. See Section 4.4 of the online appendix for details and reduced-form analysis of how other sample moments determine ΔRMSE.

12. Figure 4 in Section 4.4 of the online appendix shows a version of these two panels for all 12 sample moments that determine ΔRMSE at the country level.

data. Reduction in the bias of the estimators drives most of the changes in RMSE rather than a smaller expected variance. Bias drives 98.5 percent of RMSE on average when non-renewables are excluded (model 1), and 81 percent of bias on average when non-renewables are included (model 2).[13] Additional cross-sectional analysis examining how ΔRMSE at the country-level relates to other factors that play a role in measuring productivity (e.g., statistical capacity and the share of GDP stemming from resource rents) can be found in Section 4.4 of the online appendix.

The resulting decline in measurement error for most countries is of a magnitude that is policy-relevant. The median reduction in RMSE across the 100 countries in our sample is 6.6 basis points. This translates to an annual forecast error of roughly 0.07 percentage points when historical estimates are used to forecast average growth in the future. Applying this median value to the United States, when this forecast error is compounded over ten years (the standard CBO budget horizon) at current levels of output, forecast cumulative revenues would deviate from projections by approximately \$300 billion.[14] This forecast error is roughly one quarter of the aggregate social cost of compliance with the U.S. Clean Air Act between 1970 and 1990, one of the broadest environmental policies enacted globally to date (U.S. Environmental Protection Agency 1997).

Comparison with existing results: The magnitudes of ΔRMSE induced by including non-renewables here are similar to those produced by Brandt, Schreyer, and Zipperer (2017). The authors there find including non-renewable capital changes point estimates of annual TFP growth by an average of 2.3 basis points for a panel of 21 countries between 1986 and 2008. Manning and Barbier (2025) find adjusting the growth accounting equation to account for natural capital increases the amount of output growth unexplained by factor inputs by 87 basis points on average for countries with negative historical TFP growth and decreases it by 65 basis points on average for countries where TFP growth has been positive. Freeman, Inklaar, and Diewert (2021) use an earlier version of the CWON dataset to adjust TFP estimates to account for subsurface minerals and fossil fuels and find omitting non-renewable resources can bias TFP levels by up to 50 percent for resource-intensive economies.

Sensitivity Analysis: Section 4.5.1 of the online appendix performs a Monte Carlo analysis to examine the sensitivity of our results to alternative pairs of output elasticities for the two natural capital stocks. We recalibrate the simulation ten million times by reassigning output elasticities based on iid draws from a uniform distribution on the interval $[0, 0.1]$. The median (mean) ΔRMSE at the country level falls to 2 (28) basis points and is positive under about 30 percent of alternative parameterizations, highlighting the importance of better measurement and apportionment of income generated by natural capital. Section 4.5.2 shows that re-estimating the sample moments used to calibrate country-level data generating processes based on two partitions of the original 1996-2019 data leads to larger estimates of RMSE reductions across the board and

13. See Section 4 of the online appendix for the derivation and decomposition of the changes in RMSE in terms of changes in bias and variance.

14. See Section 5 of the online appendix for a description of the back-of-the-envelope calculation reported here. The cumulative deviation for the United States would grow to slightly under \$1.2 trillion if we were to instead apply our U.S.-specific point estimate for the change in RMSE (26 bps) when including non-renewables on the margin.

minimal qualitative changes in the set of countries for which ΔRMSE is negative.

4 Conclusion

Fifty-four years after Nordhaus and Tobin (1973), research continues to highlight new channels by which environmental change affects society and welfare. An increasing share of this research has focused on responding to high-level calls for improved welfare measures, e.g., the U.N. Secretary General's push for measures "Beyond GDP" (United Nations Statistics Division 2026). Economists have long recognized the importance of such measures, and there is a rich literature on complements to GDP that would better account for the determinants of welfare beyond market activity (Fleurbaey 2009; Jorgenson 2018).[15] This paper gives evidence that measurement of natural capital benefits governments irrespective of goals beyond GDP. When natural capital affects production, omitting it from national accounts biases estimates of economic growth. This supports the original hypothesis by Nordhaus and Tobin in 1973 that leaving things out ("treating as free things that are not really free") of national accounts "... gives the wrong signals for the directions of economic growth" in addition to affecting measurement of national welfare (Nordhaus and Tobin 1973). While measuring the environment will remain important for assessing the sustainability of development, regulatory compliance, and public health, we show here that systematically connecting changes in the environment to economic statistics will benefit existing measurement that is already used to guide policy.

15. Conte, Addicott, and Millerhaller (2025) provide micro-founded support for this position regarding the environment, demonstrating how reductions in natural capital stocks can generate real income effects for consumers.

Table 1: TFP Growth vs. Growth in Renewable Natural Capital Stocks

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Urban Land	0.301	0.303	0.205	0.215	0.149	0.060
	(0.095)	(0.098)	(0.067)	(0.075)	(0.062)	(0.118)
Timber Land	0.164	0.166	0.071	0.075	-0.003	0.026
	(0.090)	(0.075)	(0.074)	(0.071)	(0.068)	(0.104)
Agricultural Land	0.027	0.043	0.030	0.036	0.041	0.043
	(0.025)	(0.025)	(0.022)	(0.023)	(0.021)	(0.017)
Forest Cover	-0.067	-0.111	-0.005	0.012	0.083	-0.027
	(0.094)	(0.091)	(0.079)	(0.083)	(0.053)	(0.076)
Mangroves	0.122	0.067	0.357	0.327	0.318	-0.108
	(0.244)	(0.242)	(0.299)	(0.322)	(0.367)	(0.427)
Fishery Biomass	0.112	0.026	0.094	0.072	0.047	-0.104
	(0.098)	(0.077)	(0.108)	(0.101)	(0.135)	(0.108)
Hydropower	0.011	0.011	0.009	0.009	0.006	0.006
	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)
Controls for...						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.007	0.003	0.007	0.005	0.002	0.039
Bootstrap Wald	0.089	0.056	0.044	0.018	–	0.037
# Observations	2,808	2,808	2,808	2,808	2,808	2,457
# Countries	117	117	117	117	117	117

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables (Feenstra, Inklaar, and Timmer 2015). All independent variables enter as log changes. Standard errors clustered at the country level are shown in parentheses. Rows under the "Controls for ..." heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator and reports robust (as opposed to clustered) standard errors. Row "Wald" displays p-values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis (Wu 1986). No results are reported for this test in column (5) due to the idiosyncratic trend terms.

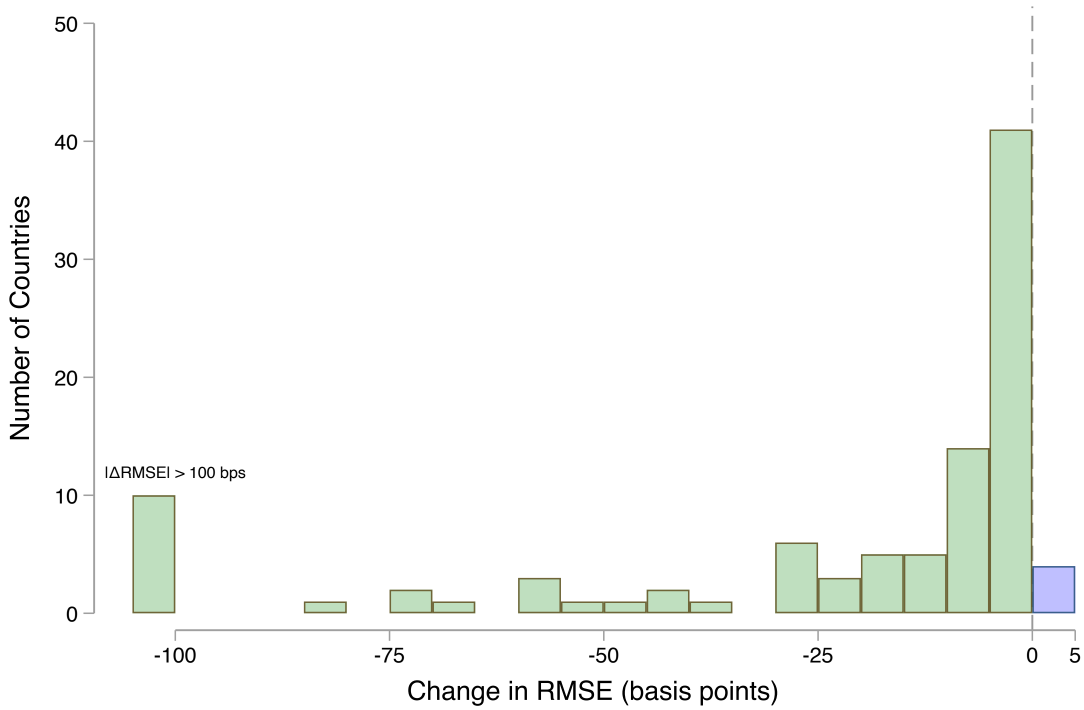

Figure 1: Histogram of the reductions in root mean squared error—ΔRMSE—when adding g_{N_1} to the Solow residual-based TFP estimator for each of 100 countries in the sample. The 10 countries with RMSE reductions larger than 100 basis points in magnitude are topcoded and added to the leftmost bin in the histogram. See Table 15 in the online appendix for a full list of country-level sample moments and changes in RMSE.

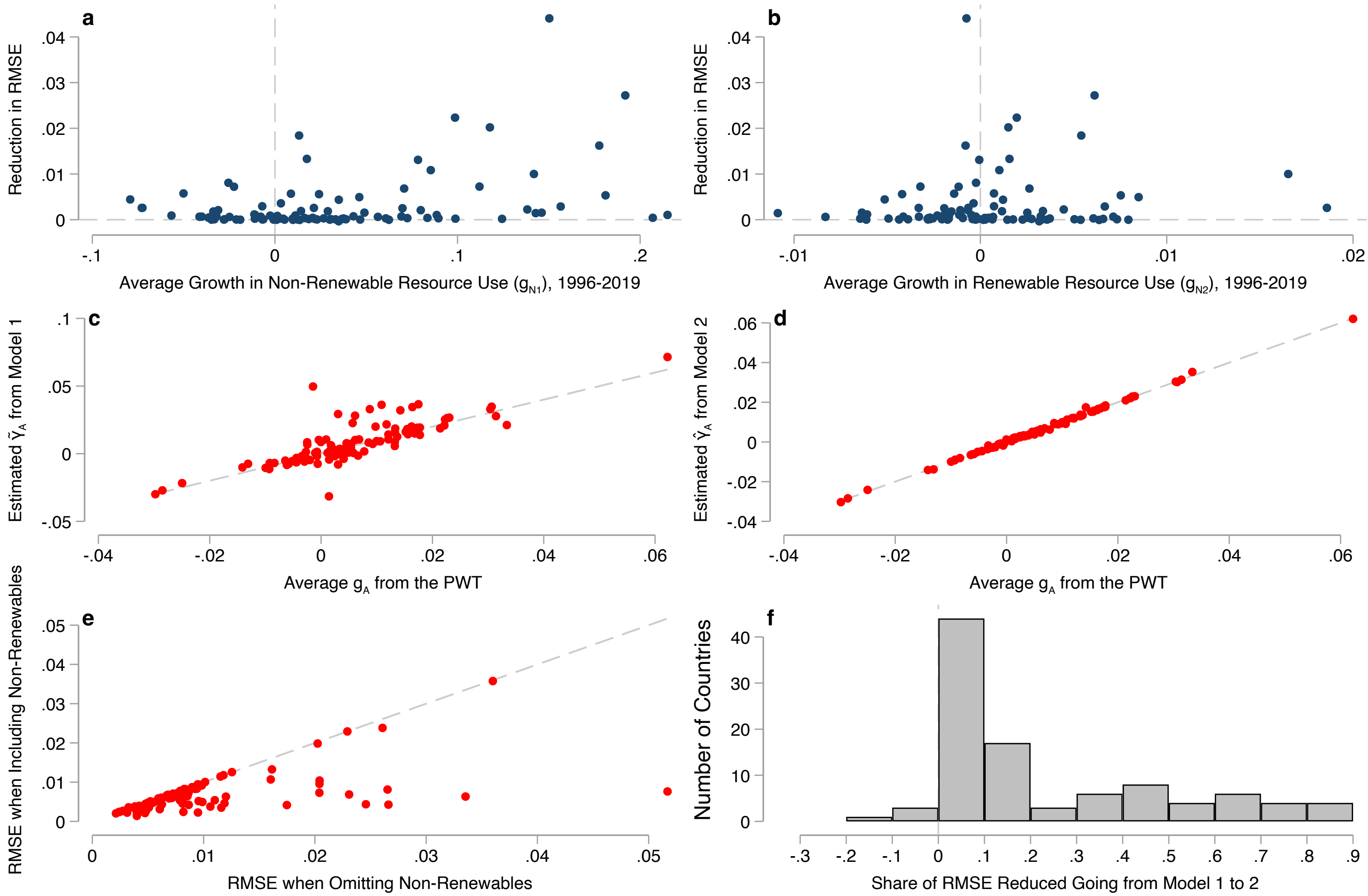

Figure 2: Panels 2a and 2b plot ΔRMSE when estimating TFP growth against average growth in non-renewable and renewable resource use at the country level between 1996 and 2019 for our panel of 100 countries. Panels 2c and 2d show point estimates of TFP growth based on historical moments from the two estimators against the true data generating process. Panels 2e and 2f show the changes in RMSE from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this change. Dashed lines in panels 2a, 2b, and 2f are plotted at $x = 0$ and $y = 0$. Dashed lines in panels 2c-2e show the 45 degree line.

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